2. SCENARIOS

Scenario A: Adaptive Meal Planning

A user logging in for the first time inputs their biometrics (age, weight, height) and specific fitness goals into the Meal Planner tab. The system instantly calculates their basal metabolic rate (BMR) and total daily energy expenditure (TDEE). Utilizing this data, the Groq LLM generates a dynamically balanced, macro-calibrated 7-day meal plan tailored perfectly to their dietary preferences and restrictions.

Scenario B: Freshness & Spoilage Checking

A home cook is unsure if a piece of produce such as fruits , vegetables or meat is still safe to eat. They upload a photo to the Freshness Detector. The local computer vision algorithms calculate surface discoloration, dark spots, and texture integrity. This metric data is sent to the AI, which returns an estimated shelf-life (e.g., "3-5 days") and concrete safety recommendations based on visual degradation.

Scenario C: Kitchen Hygiene Auditing

A restaurant manager wants to ensure compliance in their commercial kitchen. They upload a video keyframe of a prep station to the Hygiene Monitor. The multimodal AI scans the image for critical safety barrier protection—such as glove usage and hair-cap compliance—as well as workstation clutter. The system generates an objective sanitation score out of 100, highlighting areas for immediate remediation.

Scenario D: AI Chef & Multimodal Recipe Generation

A home cook wants to prepare a meal but only has a few random ingredients in their pantry. They upload a photo of the available ingredients and use voice (via Groq Whisper) or text instructions to request a "high-protein, 30-minute pasta dish." The AI Chef processes the image, transcribes the audio, and synthesizes a personalized, step-by-step culinary guide complete with macro targets, prep times, and executive chef secrets.

Scenario E: Daily Calorie Tracking & Goal Setting

A user wants to monitor their dietary progress over the course of a day. They navigate to the Global Dashboard and set a personalized daily calorie goal (e.g., 2000 kcal). As they log meals, the Gamified Calorie Quest tracker updates in real-time. Visual progress bars and milestone badges—such as "Morning Fuel" and "Goal Crushed!"—unlock dynamically, providing an engaging and accountable way to meet nutritional targets.